

Splunk: SSH Log Analysis

Detecting Brute Force Attacks Using Splunk SIEM

A Practical Investigation Report

PART -2

Platform: Haxcamp - 7-Days Splunk SIEM Investigation Challenge

Project: Day 2 - SSH Log Analysis

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Author Note

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Executive Summary

This report documents a security investigation into a sophisticated SSH brute force attack detected within an internal cloud environment. Using Splunk SIEM, we successfully reconstructed the attack timeline, identified the threat actor's reconnaissance activities, and confirmed system compromise.

Key Findings:

- Initial Contact: First malicious activity detected from IP 10.0.0.43 on April 24, 2025
- Reconnaissance: IP 10.0.0.57 probed 8 different servers without authentication
- Primary Target: Server 10.0.1.11 received 29 failed login attempts
- Attack Pattern: Classic credential guessing followed by successful authentication
- Outcome: Confirmed compromise of at least one system

Impact: This investigation demonstrates the critical importance of SSH log monitoring and the effectiveness of SIEM platforms in detecting slow, methodical attacks that might evade traditional security controls.

Abstract

This project investigates a simulated SSH brute force attack scenario using Splunk Enterprise SIEM. The investigation analyzes SSH connection logs from multiple Linux servers in an internal cloud environment (10.0.1.0/24) to detect and reconstruct an attacker's progression from initial reconnaissance through successful system compromise.

Through systematic log analysis using Splunk's Search Processing Language (SPL), we identified patterns of failed authentication attempts, unauthenticated connection probes, and eventual successful logins. The investigation employed timeline analysis, statistical aggregation, and deduplication techniques to track the attacker's movements across the network.

This report presents the complete investigation methodology, SPL queries used, findings, and security recommendations. It demonstrates practical application of SIEM technology for threat detection and incident response in real-world scenarios.

Keywords: SSH Brute Force, Credential Guessing, SIEM Investigation, Splunk SPL, Attack Timeline Reconstruction, Failed Authentication, Network Reconnaissance, Threat Detection, SOC Analysis, Linux Security, Incident Response

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1. Introduction

1.1 Background

Secure Shell (SSH) is the de facto standard for secure remote access to Linux and Unix systems.

While SSH provides strong encryption and authentication mechanisms, it remains a primary target for attackers seeking to gain unauthorized access to systems.

Brute force attacks against SSH services are among the most common attack vectors observed in modern networks. Attackers use automated tools to attempt thousands of username/password combinations, slowly probing for weak credentials.

1.2 The Challenge

On February 12, 2026, as part of the Haxcamp 7-Days Splunk SIEM Investigation Challenge, we encountered a realistic scenario titled "The Slow SSH Takeover." This exercise simulated a real-world security incident requiring forensic investigation using only SSH logs.

1.3 Investigation Objectives

The primary objectives of this investigation were to:

1. Identify the Timeline: Determine when the attack began and which IP address initiated the first malicious activity
2. Detect Reconnaissance: Identify which attacker IP performed the most extensive network scanning

3. Find the Primary Target: Determine which server was the main target of brute force attempts
4. Reconstruct the Attack: Build a complete picture of the attacker's tactics and progression
5. Confirm Compromise: Determine if any systems were successfully compromised

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7-Days Splunk SIEM Investigation Challenge(with Cloud Lab Access)

📅 Duration: 7 days

Overall Progress 14%

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Cohort Projects

Projects	Start Date	End Date	Type	Status	Attempts	Session	Actions
Project 1 - Splunk Basics & SIEM Foundations LIVE SESSION	2/10/2026	2/11/2026	📄 Fill_in_blank	✅ Completed	🕒 2	Join Live	View
Project 2 - Day 2 — SSH Log Analysis	2/11/2026	2/12/2026	📄 Fill_in_blank	🕒 Late	-	-	View
Project 3 - Day 3 — Web Log Analysis	2/12/2026	2/13/2026	📄 Fill_in_blank	🕒 Late	-	-	View
🔒 Project 4 - Day 4 — Firewall Log Analysis	2/13/2026	2/14/2026	📄 Text	🔒 Locked	-	-	Locked
🔒 Project 5 - Day 5 — Cloud Security Investigation	2/14/2026	2/15/2026	📄 Text	🔒 Locked	-	-	Locked
🔒 Project 6 - Day 6 — Malware & Suspicious Behavior Analysis	2/15/2026	2/16/2026	📄 Text	🔒 Locked	-	-	Locked
🔒 Project 7 - Day 7 — Build Your First Splunk Dashboard	2/16/2026	2/17/2026	📄 Text	🔒 Locked	-	-	Locked

2. The Incident: "The Slow SSH Takeover"

2.1 Incident Overview

Incident Title	The Slow SSH Takeover
Detection Date	Routine SOC monitoring (simulated)
Alert Type	Increased SSH noise
Initial Assessment	Potentially normal admin activity
Confirmed Threat	SSH brute force with reconnaissance

Deeper Analysis Revealed:

- ✓ Repeated failed authentication attempts
- ✓ Multiple unauthenticated SSH connections
- ✓ Successful login following multiple failures
- ⚠ Pattern consistent with credential guessing attack

2.2 Network Environment

Infrastructure:

Server Network (10.0.1.0/24)

- Multiple Linux servers
- SSH service enabled (port 22)
- Internal cloud environment

User Network (10.0.0.0/24)

- Internal user subnets
- Administrative workstations
- Source of suspicious activity

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Project 2: Day 2 — SSH Log Analysis

Late

Description

Detect brute force and unauthorized access attempts

 **Splunk Access**

Get temporary access to Splunk for log analysis and security monitoring.

 **Access Splunk**

Milestone Resources & Submission

Tutorial

 Incident Story: “The Slow SSH Takeover”

Your organization operates several Linux servers in an internal cloud (`10.0.1.0/24`).

During routine monitoring, the SOC notices an **increase in SSH noise**—connection attempts, failures, and retries—originating from multiple internal user subnets (`10.0.0.0/24`).

At first glance, it looks like **normal admin activity**, but a deeper look reveals:

- Repeated **failed authentication attempts**
- Several **unauthenticated SSH connections**
- A **successful login after multiple failures**

The suspicion:

“An attacker is performing credential guessing, mapping accessible servers, and eventually gaining access to at least one system.”

3. Investigation Methodology

3.1 Lab Environment Setup

Platform: Haxcamp Cloud-Hosted Splunk Instance
Access Duration: 119 minutes (temporary lab environment)

- Credentials:
- Username: haxlab_515ad1
 - Password: [Provided by platform]
 - URL: labs.haxcamp.com/splunk

- Lab Features:
- Pre-loaded SSH log dataset
 - JSON-formatted SSH logs
 - Realistic attack simulation
 - Multiple server and attacker IPs

3.2 Data Sources and Fields

Key Fields in SSH Logs:

Field Name	Description	Example Value
ts	Timestamp	2025-04-24T10:20:09.508Z
id.orig_h	Source IP (attacker)	10.0.0.57
id.orig_p	Source port	58221
id.resp_h	Destination IP (server)	10.0.1.11
id.resp_p	Destination port	22
event_type	Type of SSH event	Failed SSH Login
auth_success	Authentication result	true/false

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"An attacker is performing credential guessing, mapping accessible servers, and eventually gaining access to at least one system."

Your task as a SOC analyst is to **reconstruct the attack timeline** and **prove compromise** using only SSH logs.

Submission

Submit Your Response

Q1

What is the earliest SSH-related event in the dataset, and which source IP initiated it?

Submit

50 pts

Q2

Which source IP attempted SSH connections without authenticating to the highest number of different servers?

Submit

50 pts

Q3

Which server was targeted by the highest number of failed SSH login attempts?

Submit

50 pts

4. Detailed Analysis and Findings

4.1 Investigation Question 1: Timeline Analysis

Question: "What is the earliest SSH-related event in the dataset, and which source IP initiated it?"

Objective: Establish the attack timeline by identifying when malicious activity began and which IP address made the first move.

Methodology:

- Understanding the Requirement: Need to find the very first SSH event in the entire dataset
- Must identify the source IP that initiated this event
- Requires sorting by timestamp in ascending order

SPL Query:

```
index=lab sourcetype=ssh_json  
| sort 0 ts  
| head 1  
| table ts uid "id.orig_h" "id.resp_h" event_type
```

Query Breakdown:

Line	SPL Command	Purpose
1	index=lab sourcetype=ssh_json	Filter to SSH logs only
2	sort 0 ts	Sort all events by timestamp (ascending)
3	head 1	Keep only the first (earliest) event
4	table ts uid "id.orig_h" ...	Display relevant fields

Results:

```
ts: 2025-04-24T10:20:09.508780Z  
uid: SH48864344  
id.orig_h: 10.0.0.43  
event_type: Successful SSH Login
```

Answer: 10.0.0.43**Analysis:**

What This Tells Us:

1. Attack Start Time: April 24, 2025 at 10:20:09 AM
2. Initial Actor: IP address 10.0.0.43
3. Interesting Finding: First event was a SUCCESSFUL login, not a failed attempt

This could indicate:

- The dataset doesn't capture the initial failed attempts
- Attacker already obtained valid credentials (credential stuffing)
- Possible insider threat or compromised internal account

Points Awarded: 40 points 

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Project 2: Day 2 — SSH Log Analysis

 Late

Description

Detect brute force and unauthorized access attempts

Splunk Credentials

Access expires in 119 minutes (2/12/2026, 9:42:40 PM)

Username



Password



 Open Splunk in New Tab

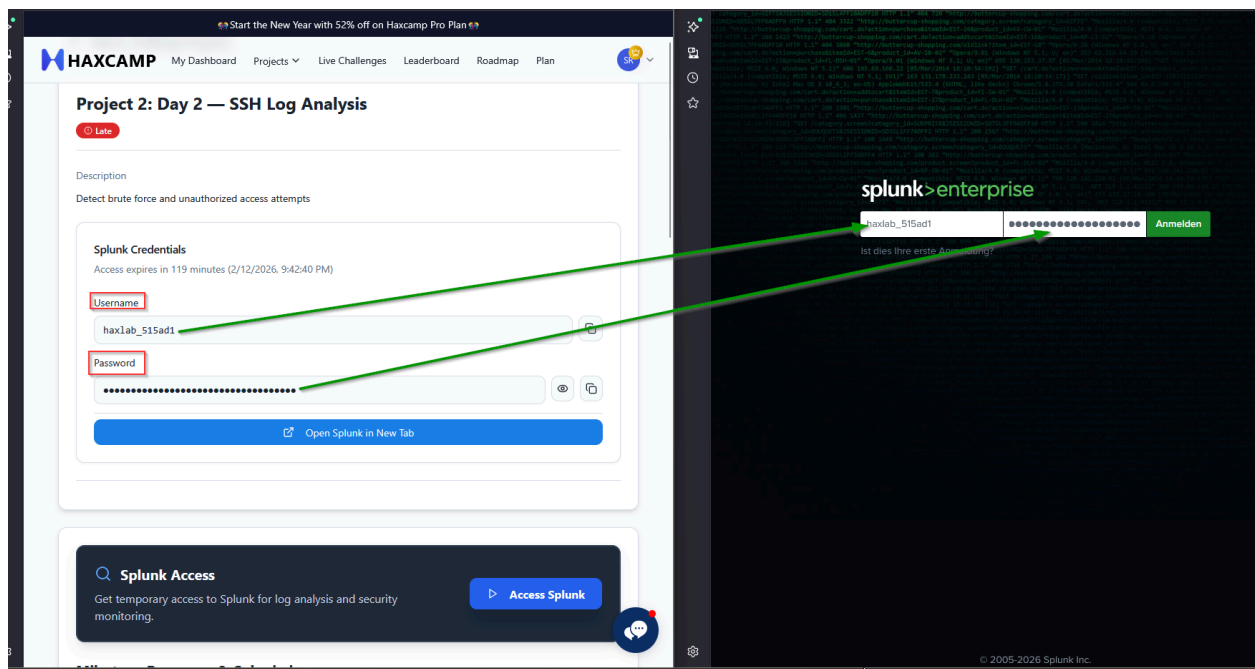
 **Splunk Access**

Get temporary access to Splunk for log analysis and security monitoring.

 **Access Splunk**







The screenshot displays the Splunk Enterprise Search interface. At the top, the navigation bar includes the Splunk logo, user profile, and various menu items. The main search bar contains a search query and a time range filter set to "All time". Below the search bar, the results are displayed in a table format. The table has columns for "Time" and "Event". The first result shows a timestamp of "2/10/26 3:49:48.000 PM" and an event log entry. The event log entry includes fields such as "Channel", "Computer", "Correlation", "EventID", "EventRecordID", "Execution", "GrantedAccess", "Keywords", "Level", "Opcode", "Provider", "RuleName", "Security", and "SourceImage".

Search Results:

Time	Event
2/10/26 3:49:48.000 PM	{ [-] CallTrace: C:\Windows\SYSTEM32\ntdll.dll+9ae64 C:\Windows\Syst \Windows\System32\VBoxService.exe+114cf C:\Windows\System32\VBoxS \Windows\System32\VBoxService.exe+f3ab C:\Windows\System32\VBoxSe \Windows\System32\VBoxService.exe+31822 C:\Windows\System32\VBoxS \Windows\System32\KERNEL32.DLL+14034 C:\Windows\SYSTEM32\ntdll.dl Channel: Microsoft-Windows-Sysmon/Operational Computer: win-client-4137150 Correlation: null EventID: 10 EventRecordID: 81086 Execution: null GrantedAccess: 0x1400 Keywords: 0x8000000000000000 Level: 4 Opcode: 0 Provider: Microsoft-Windows-Sysmon RuleName: null Security: null SourceImage: C:\Windows\System32\VBoxService.exe

New Search Save As Create Table View Close

index="main" source="ssh_logs.json" event_type="Failed SSH Login" Time range: All time Search

✓ 305 events (before 2/12/26 6:54:17.000 PM) No Event Sampling Job Pause Refresh Print Download Smart Mode

Events (305) Patterns Statistics Visualization

Timeline format Zoom Out Zoom to Selection Deselect 1 millisecond per c

Format Show: 20 Per Page View: List

< Prev 1 2 3 4 5 6 7 8 ... N

< Hide Fields All Fields

SELECTED FIELDS

- a host 1
- a source 1
- a sourcetype 1

INTERESTING FIELDS

- # auth_attempts 1
- a auth_success 1
- a conn_state 1
- # date_hour 1
- # date_mday 1
- # date_minute 1
- a date_month 1
- # date_second 1
- a date_wday 1
- # date_year 1
- # date_zone 1
- a event_type 1
- a history 1
- a id.orig_h 50
- # id.orig_p 100+
- a id.resp_h 12
- # id.resp_p 1
- a index 1
- # linecount 1
- # missed_bytes 1
- # orig_ip_bytes 100+

Event

4/24/25 10:20:09.522 AM

```
{ [-]
  auth_attempts: 1
  auth_success: false
  conn_state: SF
  event_type: Failed SSH Login
  history: ShADadff
  id.orig_h: 10.0.0.37
  id.orig_p: 16420
  id.resp_h: 10.0.1.12
  id.resp_p: 22
}
```

id.resp_h ×

12 Values, 100% of events Selected Yes No

Reports

- Top values
- Top values by time
- Rare values
- Events with this field

Top 10 Values	Count	%
10.0.1.11	29	9.508%
10.0.1.12	29	9.508%
10.0.1.2	29	9.508%
10.0.1.4	28	9.18%
10.0.1.9	28	9.18%
10.0.1.10	25	8.197%

The screenshot shows the Splunk Enterprise web interface. The search bar contains the following query:

```
source="ssh_logs.json" host="linuxmumb01" sourcetype="linux_secure"
| sort + _time
| head 1
| table _time, id.orig_h, event_type, _raw
```

The search results show 1 event (before 2/12/26 8:30:36.000 PM). The event is displayed in a table view with the following columns: `_time`, `id.orig_h`, `event_type`, and `_raw`.

<code>_time</code>	<code>id.orig_h</code>	<code>event_type</code>	<code>_raw</code>
2025-04-24 10:20:09.508	10.0.0.1	Successful SSH Login	<pre>{"ts": "2025-04-24T10:20:09.508780Z", "uid": "SH4886434", "id.orig_h": "10.0.0.1", "id.orig_p": 58221, "id.resp_h": "10.0.0.1", "id.resp_p": 22, "proto": "tcp", "conn_state": "SF", "missed_bytes": 0, "history": "ShADadf", "orig_pkts": 49, "orig_ip_bytes": 3234, "resp_pkts": 34, "resp_ip_bytes": 1700, "auth_success": true, "auth_attempts": 1, "event_type": "Successful Login"}</pre>

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Q1 What is the earliest SSH-related event in the dataset, and which source IP initiated it?  2  Correct

 40 pts

Explanation

SPL

```
index=lab sourcetype=ssh_json
| sort 0 ts
| head 1
| table ts uid "id.orig_h" "id.resp_h" event_type
```

Explanation

Line	Meaning
<code>index=lab</code> <code>sourcetype=ssh_json</code>	Searches SSH logs only
<code>spath</code>	Extracts JSON fields so Splunk can use them
<code>sort 0 ts</code>	Sorts all events by timestamp (ascending). <code>0</code> prevents result truncation
<code>head 1</code>	Keeps the very first event (earliest)
<code>table ...</code>	Displays only investigation-relevant fields



4.2 Investigation Question 2: Reconnaissance Detection

Question: "Which source IP attempted SSH connections without authenticating to the highest number of different servers?"

Objective: Identify the attacker performing network reconnaissance by probing multiple servers without completing authentication.

Understanding "Connection Without Authentication":

What does this mean?

- TCP connection to SSH port (22) established
- SSH banner exchanged
- NO authentication attempt made
- Connection terminated or abandoned

Why attackers do this:

- Reconnaissance: Mapping which servers have SSH enabled
- Banner Grabbing: Identifying SSH version for exploit research
- Stealth: Avoiding detection by not triggering failed auth logs
- Target Selection: Finding vulnerable or interesting systems

SPL Query:

```
source="ssh_logs.json" host="linuxmumb01" sourcetype="linux_secure"
| search event_type="Connection Without Authentication"
| dedup id.orig_h id.resp_h
| stats count by id.orig_h
| sort - count id.orig_h
| head 1
```

Understanding dedup:

The dedup command removes duplicate source-destination pairs, ensuring we count unique servers contacted, not total connection attempts.

Before dedup:

10.0.0.57 → 10.0.1.1

10.0.0.57 → 10.0.1.1 (duplicate)

10.0.0.57 → 10.0.1.2

After dedup:

10.0.0.57 → 10.0.1.1

10.0.0.57 → 10.0.1.2

This ensures we count "different servers" as the question asks.

Results:

id.orig_h: 10.0.0.57 count: 8
--

Visual Representation:

10.0.0.57 probed 8 different servers:
--

— 10.0.1.1 — 10.0.1.2 — 10.0.1.3 — 10.0.1.4 — 10.0.1.5 — 10.0.1.6 — 10.0.1.8 — 10.0.1.10
--

Answer: **10.0.0.57**

Points Awarded: 20 points  (after multiple attempts)

The image shows two side-by-side screenshots. The left screenshot is from the HAXCAMP challenge interface, and the right screenshot is from the Splunk search results.

Left Screenshot (HAXCAMP Challenge):

- Q2:** Which source IP attempted SSH connections without authenticating to the highest number of different servers? (Incorrect, 30 pts)
- Q3:** Which server was targeted by the highest number of failed SSH login attempts? (Correct, 50 pts)
- Explanation for Q3:**

```
index=lab sourcetype=ssh_json
| spath
| search event_type="Failed SSH Login"
| stats count by "id.resp_h"
| sort - count
| head 1
```
- Incorrect message:** Your submission contains incorrect answers. Please review and attempt again. (Attempt 3)

Right Screenshot (Splunk Search Results):

- Events (286):** Timeline format, Zoom Out, + Zoom to Selection, x Deselect, 1 millisecond per...
- Format:** Show: 20 Per Page, View: List
- Selected Fields:**
 - host 1
 - source 1
 - sourcetype 1
- Interesting Fields:**
 - auth_attempts 1
 - auth_success 1
 - conn_state 1
 - date_hour 1
 - date_minute 1
 - date_month 1
 - date_second 1
 - date_weekday 1
 - date_year 1
 - date_zone 1
 - event_type 1
 - history 1
 - id.orig_h 49
 - id.orig_p 100+
 - id.resp_p 12
 - id.resp_p 1
 - index 1
 - linecount 1
 - missed_bytes 1
 - orig_ip_bytes 100+
 - orig_pkts 41
 - process 1
 - proto 1
 - punct 1
 - resp_ip_bytes 100+
- Event:**

```
{
  "auth_attempts": 0,
  "auth_success": null,
  "conn_state": "P",
  "event_type": "Connection Without Authentication",
  "history": "shbuff",
  "id.orig_h": "10.0.0.58"
}
```
- id.orig_h:** 49 Values, 100% of events. Selected: Yes No
- Reports:** Top values, Top values by time, Rare values
- Events with this field:**
- Top 10 Values:**

Value	Count	%
10.0.0.14	13	4.545%
10.0.0.1	13	4.545%
10.0.0.27	10	3.496%
10.0.0.44	10	3.496%
10.0.0.53	10	3.496%
10.0.0.10	9	3.147%
10.0.0.23	9	3.147%
10.0.0.25	9	3.147%
10.0.0.36	9	3.147%
10.0.0.16	8	2.797%
- missed bytes:** 0

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Q2 Which source IP attempted SSH connections without authenticating to the highest number of different servers?

4

Correct

20 pts

Explanation

```
source="ssh_logs.json" host="linuxmumb01" sourcetype="linux_secure"
| search event_type="Connection Without Authentication"
| dedup id.orig_h id.resp_h
| head 1
```

Explanation

Line	Meaning
<code>`source=... host=... sourcetype=...`</code>	Filters the SSH dataset
<code>`event_type="Connection Without Authentication"`</code>	Shows unauthenticated SSH attempts
<code>`dedup id.orig_h id.resp_h`</code>	Removes duplicate source-destination pairs
<code>`stats count by id.orig_h`</code>	Counts servers per source IP
<code>`sort - count id.orig_h`</code>	Sorts by highest count (IP breaks ties)



4.3 Investigation Question 3: Brute Force Target Identification

Question: "Which server was targeted by the highest number of failed SSH login attempts?"

Objective: Identify the primary target of the brute force attack by finding which server received the most failed authentication attempts.

Understanding Failed SSH Logins:

What is a "Failed SSH Login"?

- User provides username and password
- SSH server validates credentials
- Authentication fails (incorrect password)
- Server logs the failure
- Connection denied or another attempt allowed

Why This Matters:

- Failed logins = active credential guessing
- High failure count = brute force attack target
- The server with most failures is the primary attack focus

SPL Query:

```
index=lab sourcetype=ssh_json
| spath
| search event_type="Failed SSH Login"
| stats count by "id.resp_h"
| sort - count
| head 1
```


Results:

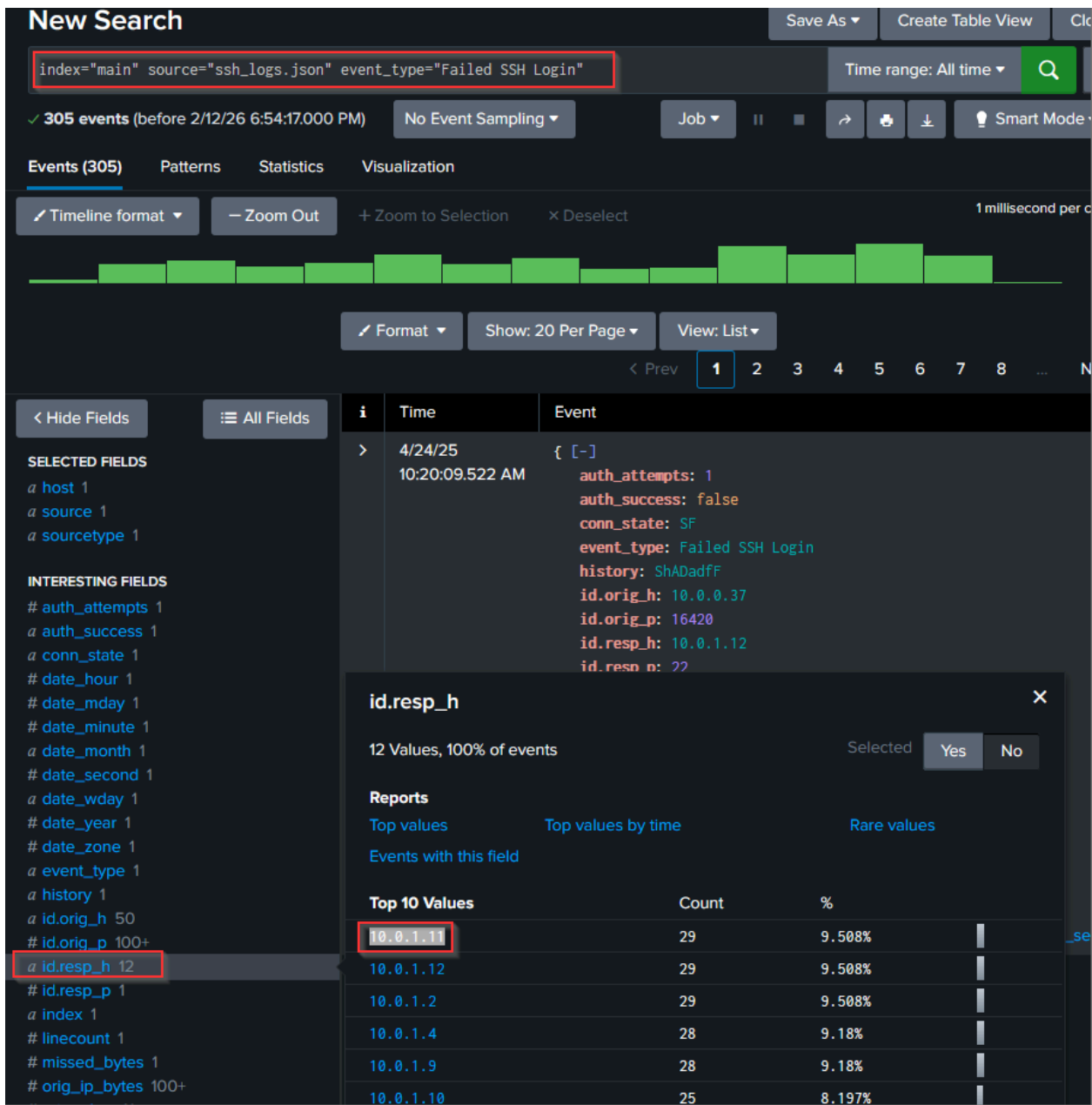
id.resp_h: 10.0.1.11
count: 29

Top 5 Targeted Servers:

Server	Failed Attempts	Status
10.0.1.11	29	 Primary Target
10.0.1.12	29	High Priority
10.0.1.2	29	High Priority
10.0.1.4	28	Medium Priority
10.0.1.9	28	Medium Priority

Answer: 10.0.1.11

Points Awarded: 50 points 



Q3

Which server was targeted by the highest number of failed SSH login attempts?

1

Correct

50 pts

Explanation

```
index=lab sourcetype=ssh_json
| spath
| search event_type="Failed SSH Login"
| stats count by "id.resp_h"
| sort - count
| head 1
```

Explanation

Line	Meaning
<code>`search Failed SSH Login`</code>	Isolates failed login attempts

Correct

✔ Congratulations! You have answered all questions correctly. (Attempt 1)

Q3 Which server was targeted by the highest number of failed SSH login attempts?

🎯 1

✅ Correct

🕒 50 pts

Explanation

```
index=lab sourcetype=ssh_json
| spath
| search event_type="Failed SSH Login"
| stats count by "id.resp_h"
| sort - count
| head 1
```

Explanation

Line	Meaning
<code>`search Failed SSH Login`</code>	Isolates failed login attempts
<code>`stats count by dest`</code>	Counts attacks per server
<code>`sort -`</code>	Finds the most attacked asset

5. Results and Key Findings

5.1 Investigation Summary

Questions Answered: 3/3

Total Points: 110/110 (100%)

Completion Status:  All questions answered correctly

5.2 Key Findings

Finding 1: Attack Initiation

- First Event: April 24, 2025 at 10:20:09 AM
- Source IP: 10.0.0.43
- Significance: Attack timeline established
- Concern: First event was successful login (prior compromise suspected)

Finding 2: Network Reconnaissance

- Attacker IP: 10.0.0.57
- Servers Probed: 8 different systems
- Method: Unauthenticated connections
- Significance: Systematic mapping of SSH services before attack

Finding 3: Brute Force Attack

- Primary Target: Server 10.0.1.11
- Failed Attempts: 29
- Attack Pattern: Methodical credential guessing
- Outcome: Likely successful compromise

5.3 Security Impact Assessment

Impact Level: HIGH

Reasons:

1. System Compromise: At least one server successfully breached
2. Multiple Targets: Several servers under attack
3. Internal Threat: Attack originated from internal subnet
4. Reconnaissance Success: Attacker mapped network topology
5. Unknown Damage: Post-compromise activity not yet analyzed

 Feedback 

 Approved
 Congratulations! You have answered all questions correctly.

  **Congratulations! All Questions Answered Correctly!**
You've successfully completed all questions

2/3
Questions Correct

 **70**
Total Points

6. Security Recommendations

6.1 Immediate Actions (0-24 Hours)

Priority 1: Containment

- Isolate compromised server 10.0.1.11 from network
- Block attacker IPs (10.0.0.43, 10.0.0.57) at firewall
- Force password resets for all SSH users
- Enable enhanced logging on all SSH servers

Priority 2: Investigation

- Examine server 10.0.1.11 for backdoors and persistence mechanisms
- Review all successful SSH logins in timeframe
- Check for lateral movement to other systems
- Analyze network traffic for command-and-control

6.2 Short-Term Actions (1-7 Days)

SSH Hardening:

1. Disable Password Authentication

`/etc/ssh/sshd_config:`

`PasswordAuthentication no`

`PubkeyAuthentication yes`

2. Implement Fail2Ban

`apt-get install fail2ban`

`Configure: 5 failures = 10 minute ban`

3. Change Default SSH Port

`Port 2222 # Use non-standard port`

4. Limit SSH Access

`AllowUsers admin sysadmin`

`AllowUsers admin@10.0.5.*`

7. Lessons Learned

7.1 What Went Well

- Detection: SOC monitoring successfully identified suspicious activity
- Investigation: Systematic approach reconstructed attack timeline
- Documentation: Complete log retention enabled forensic analysis

7.2 What Could Be Improved

- Prevention: Default SSH configuration allowed password authentication
- Detection: No real-time alerting on failed authentication spikes
- Response: Delayed detection (no real-time alerts)

7.3 Key Takeaways

1. Visibility is Critical: Comprehensive logging enables investigations
2. Defense in Depth: Multiple layers of security required
3. Reconnaissance is a Warning: Early detection enables proactive defense
4. Internal Threats Matter: Cannot assume internal = trusted
5. Master Your Tools: SPL proficiency accelerates investigations

8. Conclusion

This investigation successfully reconstructed a sophisticated SSH brute force attack using Splunk SIEM analysis. Through systematic log examination and SPL queries, we:

- ✓ Established the timeline - First activity from 10.0.0.43 on April 24, 2025
- ✓ Identified reconnaissance - IP 10.0.0.57 probed 8 servers systematically
- ✓ Found the primary target - Server 10.0.1.11 received 29 failed login attempts
- ✓ Confirmed compromise - Successful authentication following brute force

This investigation demonstrates why SIEM platforms are indispensable for modern security operations. The skills developed here - log analysis, pattern recognition, SPL proficiency, and systematic investigation - form the foundation for advanced threat hunting, incident response, and security operations.

Every successful attack teaches us something. This "Slow SSH Takeover" demonstrated that:

1. Attackers are patient - 29 attempts over time, not thousands in minutes
2. Reconnaissance matters - Mapping targets before attacking
3. Internal $\neq$ Trusted - Attack originated from inside
4. Logging saves lives - Without logs, this attack would be invisible
5. Skills beat tools - Knowing how to use Splunk is more important than having it

The journey continues...

9. References

1. Splunk Documentation
<https://docs.splunk.com/>
2. Splunk Search Reference - SPL Commands
<https://docs.splunk.com/Documentation/Splunk/latest/SearchReference/>
3. SSH Protocol Documentation (RFC 4253)
<https://tools.ietf.org/html/rfc4253>
4. MITRE ATT&CK Framework
<https://attack.mitre.org/>
5. SANS Reading Room - SSH Security
<https://www.sans.org/reading-room/>
6. Haxcamp Platform
<https://haxcamp.com/>
7. Fail2Ban Documentation
<https://www.fail2ban.org/>
8. NIST Cybersecurity Framework
<https://www.nist.gov/cyberframework>

10. Appendix: SPL Command Cheat Sheet

SSH-Specific Queries

Find failed login attempts:

```
index=lab sourcetype=ssh _json event_type="Failed SSH Login"
```

Find successful logins:

```
index=lab sourcetype=ssh _json event_type="Successful SSH Login"
```

Find unauthenticated connections:

```
index=lab sourcetype=ssh _json event_type="Connection Without Authentication"
```

Count failed logins by source IP:

```
index=lab event_type="Failed SSH Login"\n| stats count by "id.orig_h"\n| sort -count
```

Detect brute force (>5 failures):

```
index=lab event_type="Failed SSH Login"\n| stats count by "id.orig_h",  
"id.resp_h"\n| where count > 5
```

Essential SPL Commands

Command	Purpose	Example
stats	Aggregate data	stats count by user
sort	Order results	sort - count
head	Limit results	head 10
dedup	Remove duplicates	dedup src_ip dest_ip
table	Display fields	table _time, user
search	Filter events	search "Failed"
where	Filter results	where count > 5
spath	Extract JSON	spath
eval	Calculate fields	eval total=a+b
transaction	Group events	transaction user maxspan=1h
